



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO R.K. FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

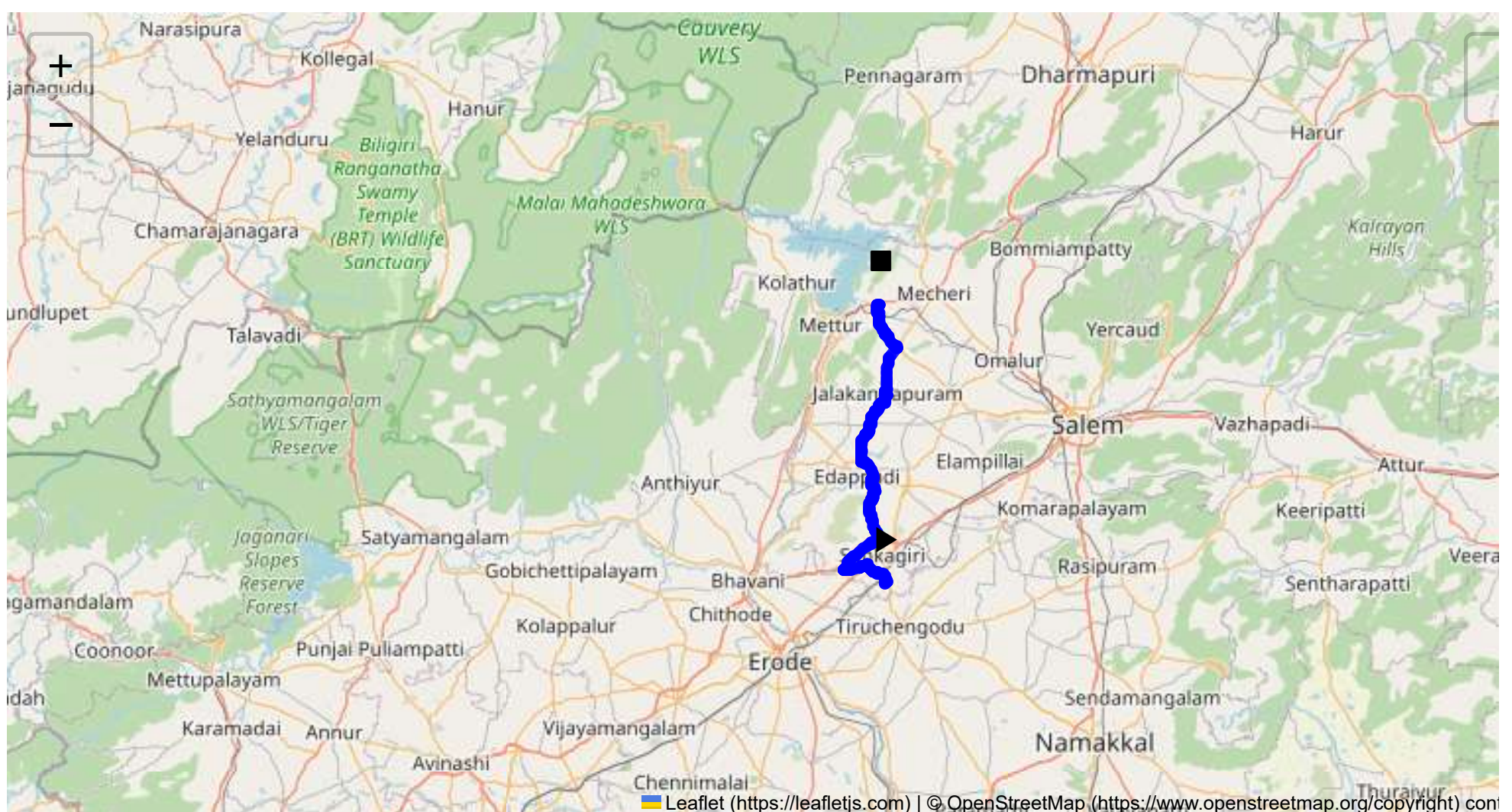
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 58.09 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.819647, 77.865112)



Welcome to the Journey Risk Management Study

- Overview of the Route Map:** The route begins at Sangagiri in Tamil Nadu, passes through Puthur, and traverses the Bhavani Main Road before continuing onto Idappadi Main Road and Idappadi Bypass, finally reaching Salem. This route spans approximately 58 kilometers and is characterized by a mix of highways and local roads, with varying traffic conditions.

2. **Typical Weather Conditions:** Tamil Nadu generally experiences a tropical climate. The region sees hot weather from March to June, with temperatures potentially creating heat stress for drivers and affecting road surfaces. The monsoon season, from October to December, can bring heavy rain, leading to waterlogging, reduced visibility, and slippery roads on certain segments.
3. **Traffic Patterns:** Traffic congestion can be expected in urban stretches, particularly during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) peak hours. The sections near major townships like Idappadi and Salem are especially prone to traffic build-up. Rural sections may have less congestion but can experience farm vehicle traffic.
4. **Assessment of Road Quality:** The main highways are generally maintained well but secondary roads, especially around rural areas, might present challenges such as potholes or uneven surfaces. Regular maintenance is less frequent on local roads, so vigilance is required.
5. **Alternative Routes:** In case of emergencies, an alternate route could involve taking the Salem-Kochi Highway (NH544) from Sangagiri directly to Salem, bypassing some of the rural roads, though this might increase travel time.
6. **Local Regulations on Hazardous Material Transport:** Particular attention should be paid to regulations about transit times for hazardous materials, usually restricted during peak traffic times. Permits are mandatory, and transport through urban zones might be regulated to certain times of the day.
7. **Historical Incidents:** While there is no direct record of specific incidents along this route, roads in Tamil Nadu do witness accidents involving heavy vehicles, often attributed to overloading, sharp curves, and sudden weather changes.
8. **Environmental Considerations:** The route traverses through some sensitive agricultural lands. Care must be taken to prevent spillage of hazardous materials, especially in proximity to water bodies to avoid contamination.
9. **Communication Coverage:** Generally, mobile network coverage is reliable in urban and semi-urban areas but may have gaps in more rural sections like near Puthur and Kollipannai. Awareness of potential communication black spots is important for ensuring safety.
10. **Estimated Emergency Response Times:** Response times vary with urban areas like Salem offering quicker emergency services in potentially less than 30 minutes. However, in rural areas and during peak traffic, it could extend to an hour or more.
11. **Overall Summary of Risk Assessment:** The route poses moderate risks mainly due to variable road conditions, traffic congestion in urban sections, and weather conditions affecting road safety. However, adherence to regulations and planning for alternatives can mitigate these risks effectively. Efficient communication planning and knowledge of local emergency services increase preparedness for incidents. Overall, the route can be navigated safely with proper caution and preparation.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43961, 77.87341	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	High	11.44898, 77.87410	15 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	High	11.45352, 77.85698	15 KM/Hr
8	Turn	Medium	11.45847, 77.85140	30 KM/Hr
9	Turn	Medium	11.45898, 77.85135	30 KM/Hr
10	Turn	Medium	11.46303, 77.84945	30 KM/Hr
11	Turn	Medium	11.46314, 77.84928	30 KM/Hr
12	Turn	Medium	11.45543, 77.81740	30 KM/Hr
13	Turn	Medium	11.45563, 77.81708	30 KM/Hr
14	Turn	High	11.45631, 77.81668	15 KM/Hr
15	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
16	Turn	Medium	11.49490, 77.86661	30 KM/Hr
17	Turn	Medium	11.50874, 77.86045	30 KM/Hr
18	Turn	Medium	11.52561, 77.85522	30 KM/Hr
19	Turn	Medium	11.54750, 77.85273	30 KM/Hr
20	Turn	Medium	11.55039, 77.85511	30 KM/Hr
21	Turn	Medium	11.55481, 77.85639	30 KM/Hr
22	Turn	High	11.57893, 77.85430	15 KM/Hr
23	Turn	Medium	11.57903, 77.85441	30 KM/Hr
24	Turn	Medium	11.58022, 77.85744	30 KM/Hr
25	Turn	Medium	11.60722, 77.84029	30 KM/Hr
26	Turn	Medium	11.60967, 77.84162	30 KM/Hr
27	Turn	Medium	11.62076, 77.84105	30 KM/Hr
28	Turn	Medium	11.63050, 77.84076	30 KM/Hr
29	Turn	Medium	11.64059, 77.84604	30 KM/Hr
30	Turn	Medium	11.64529, 77.85118	30 KM/Hr
31	Turn	Medium	11.65471, 77.85326	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
32	Turn	Medium	11.66461, 77.85598	30 KM/Hr
33	Turn	Medium	11.66601, 77.85603	30 KM/Hr
34	Turn	Medium	11.67325, 77.86052	30 KM/Hr
35	Turn	Medium	11.67371, 77.86227	30 KM/Hr
36	Turn	High	11.68489, 77.86891	15 KM/Hr
37	Turn	Medium	11.68468, 77.87272	30 KM/Hr
38	Turn	Medium	11.69294, 77.87447	30 KM/Hr
39	Turn	High	11.69472, 77.87587	15 KM/Hr
40	Turn	High	11.69493, 77.87544	15 KM/Hr
41	Turn	High	11.69510, 77.87550	15 KM/Hr
42	Turn	High	11.69552, 77.87441	15 KM/Hr
43	Turn	Medium	11.69912, 77.87474	30 KM/Hr
44	Turn	Medium	11.69938, 77.87502	30 KM/Hr
45	Turn	Medium	11.69999, 77.87509	30 KM/Hr
46	Turn	Medium	11.76005, 77.89017	30 KM/Hr
47	Turn	High	11.76020, 77.89061	15 KM/Hr
48	Turn	High	11.76075, 77.89059	15 KM/Hr
49	Turn	Medium	11.76235, 77.88747	30 KM/Hr
50	Turn	Medium	11.76254, 77.88738	30 KM/Hr
51	Turn	High	11.76370, 77.88757	15 KM/Hr
52	Turn	Medium	11.76466, 77.88539	30 KM/Hr
53	Turn	Medium	11.77813, 77.87585	30 KM/Hr
54	Turn	Medium	11.77996, 77.87351	30 KM/Hr
55	Turn	Medium	11.78162, 77.87363	30 KM/Hr
56	Turn	High	11.81084, 77.86218	15 KM/Hr
57	Turn	High	11.81583, 77.86408	15 KM/Hr
58	Turn	Medium	11.81693, 77.86352	30 KM/Hr
59	Turn	High	11.81971, 77.86457	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45563, 77.81708



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49390, 77.86748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49490, 77.86661



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50874, 77.86045



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.52561, 77.85522



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54750, 77.85273



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55039, 77.85511



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55481, 77.85639



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57893, 77.85430



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57903, 77.85441



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58022, 77.85744



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60722, 77.84029



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60967, 77.84162



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62076, 77.84105



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63050, 77.84076



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64059, 77.84604



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64529, 77.85118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65471, 77.85326



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66461, 77.85598



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66601, 77.85603



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67325, 77.86052



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67371, 77.86227



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68489, 77.86891



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68468, 77.87272



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69294, 77.87447



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69472, 77.87587



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69493, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69510, 77.87550



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69552, 77.87441



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69912, 77.87474



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69938, 77.87502



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69999, 77.87509



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76005, 77.89017



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.76020, 77.89061



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.76075, 77.89059



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76235, 77.88747



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76254, 77.88738



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76370, 77.88757



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76466, 77.88539



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77813, 77.87585



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77996, 77.87351



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78162, 77.87363



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.81084, 77.86218



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.81583, 77.86408



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81693, 77.86352



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.81971, 77.86457
